



DEEFAKE DETECTION USING ARTIFICIAL INTELLIGENCE

A Spatio-Temporal Hybrid Deep Learning Approach (ResNeXt50 + LSTM)

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1. THE DEEFAKE PROBLEM

Deepfakes synthesize hyper-realistic human facial replacements in video feeds using generative deep learning autoencoders and diffusion pipelines.

Creation Tools: FaceSwap • DeepFaceLab • Faceit • GANs / Latent Diffusion

GENERATIVE SYNTHESIS THREAT VECTORS & FORENSIC ARTIFACTS			
FACIAL RE-ENACTMENT	DEEP FACE-SWAP	LIP SYNCHRONIZATION	DIFFUSION SYNTHESIS
Generators: FaceSwap / F004s • Seamless blending • Low thermal optical jitter • Forensic Flaw: Missing temporal context	Generators: DeepFaceLab / Deep • Boundary artifacts • Poisson blending boundary seams • Forensic Flaw: Color disparity & pixel drift	Generators: Wav2Vec / Audio2Face • Lip-sync errors • Phoneme-to-visual desynchronization • Forensic Flaw: Teeth misalignment & winking	Generators: Stable Diffusion / Flux • Stochastic artifacts • High-frequency background halo • Forensic Flaw: Asymmetric contour line grids

Classification Taxonomy: Re-enactment, Face-Swap, Lip-Sync, and Latent Diffusion Flaws.

Fake News: Viral synthetic hoaxes & disinformation

Financial Fraud: Executive voice/video spoofing

Election Sabotage: Politician deepfake impersonation

Identity Theft: Defamation, non-consensual imagery

Single-frame CNNs analyze isolated stills, ignoring vital temporal continuity like optical flow, blinking jitter, and landmark coherence. Our **Spatio-Temporal Framework** captures both spatial and motion cues for robust automated defense.

+900%

Annual Surge in Malicious Synthetic Media

<54%

Human Visual Inspection Accuracy

2. FORENSIC VISUAL EVIDENCE

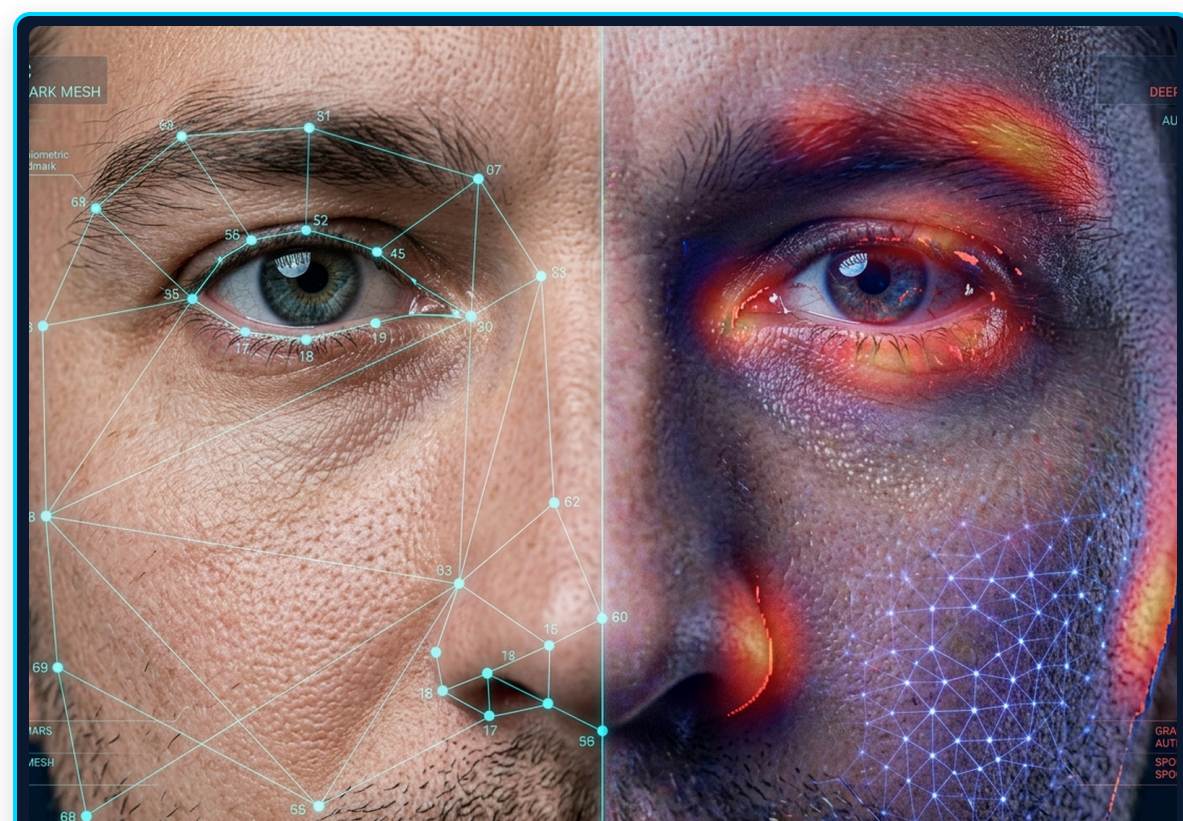


Figure 1: High-Resolution Forensic Feature Dissection & Grad-CAM Localization.
Authentic Biological Tissue & 68-Point Biometric Facial Landmark Mesh.
Generative Tamper Artifacts with High-Temperature Thermal Activation Trace.

Biometric Cue	Real Video	Deepfake Tampered
Skin Boundary	Seamless pore continuity	Blending boundary artifacts
Eye Reflection	Specular corneal coherence	Asymmetric reflection error
Landmark Grid	Smooth head kinematics	Inter-frame micro-jitter
Pupil Dynamics	Bilateral consensual response	Uncorrelated dilation jitter

Forensic Diagnostic: Grad-CAM thermal gradient analysis detects high-frequency energy concentration along boundary contours where GAN face-swaps fail seamless Poisson blending.

3. CORE OBJECTIVES & CORPUS

- Spatio-Temporal Fusion:** Couple ResNeXt-50 grouped convolutions with LSTM recurrent cells to capture micro-flickering anomalies.
- Explainability (XAI):** Generate Class Activation Maps (CAM) identifying tampered facial zones for judicial evidentiary trust.
- Generalization:** Evaluated across unseen actor identities to prevent synthetic facial overfitting.
- Sub-2s Latency:** Real-time video processing designed for scalable automated screening.

MULTI-BENCHMARK TRAINING CORPUS (23,186 VIDEOS)	
DFDC (Meta AI): 18,200 Balanced Clips (78.5%)	FaceForensics++: 3,000 High-Res Clips (12.9%)
Celeb-DF v2: 1,986 Realistic Videos (8.6%)	Data Balance: 50% Authentic / 50% Manipulated

Hardware: NVIDIA RTX 4090 / 64GB VRAM | Frameworks: PyTorch 2.0 • OpenCV 4.10

9. SOTA BENCHMARK COMPARISON

Model	Backbone	Temporal	Accuracy	AUC
MesoNet-4	Shallow CNN	None	70.4%	0.742
ResNet-50	Residual CNN	None	78.4%	0.825
XceptionNet	Depthwise Sep	None	83.2%	0.881
EfficientNet	Compound Scaled	None	84.6%	0.895
Proposed Hybrid	ResNeXt-50	Bi-LSTM	87.0%	0.924

Comparative Finding: Our proposed spatio-temporal architecture outperforms static CNN baselines by +3.8% to +16.6% across multi-dataset benchmarks.

12. APPLICATIONS & REAL-WORLD IMPACT

CYBER FORENSICS	DIGITAL KYC / BANKING	MEDIA & NEWSROOMS	SOCIAL MEDIA PLATFORMS
• Primary Objective: Court-admissible tamper reports • System Features: Cryptographic SHA-256 audit log	• Primary Objective: Face biometric verification • System Features: Anti-spoofing liveness shield	• Primary Objective: Citizen journalist feed authentication • System Features: Automated deepfake debunking	• Primary Objective: Real-time API stream filtering • System Features: Misinformation moderation bot

- Cyber Forensics:** Court-admissible SHA-256 evidentiary tamper detection reports.
- Digital KYC & Banking:** Anti-spoofing in remote video customer authentication.
- Newsroom Authentication:** Real-time verification of citizen journalist video feeds.
- Social Media Integrity:** Automated stream flagging of malicious synthetic clips.

4. DUAL-FLOW SYSTEM ARCHITECTURE

Training Flow: 23K Videos → Face Crop → **Inference Flow:** Video Upload → ResNeXt+LSTM → Verdict + Heatmap

CLIENT-SIDE WEB INTERFACE & SCREENING

Django 5.x Backend • Bootstrap 5 UI • Client-Side face-api.js 68-Landmark Mesh
Input Stream: MP4 / AVI Video → HTTP POST / Multipart Upload

UNIFORM TEMPORAL FRAME EXTRACTION

OpenCV cv2.VideoCapture • 20 Equidistant Temporal Frames Sampled (30 FPS)
Output Tensor: [Batch, T=20 Frames, H=1080, W=1920, C=3]

BIOMETRIC LOCALIZATION & CROPPING

dlib / face_recognition • 68 Landmarks • Affine Warp • 112×112 px Normalization
Normalized Face Crops: [Batch, T=20, C=3, H=112, W=112] (ImageNet Norm)

ResNeXt-50 (32×4d) SPATIAL FEATURE BACKBONE

Grouped Convolutions (Cardinality=32, Bottleneck Width=4) • Pretrained Weights
Spatial Embedding Vectors: [Batch, T=20, Feature_Dim=2048]

LSTM TEMPORAL SEQUENCE MODELING

Recurrent Memory Cells (512 Hidden Units) • Inter-Frame Optical Flow & Blinking Tracker
Sequence Hidden State: $h_t = \text{LSTM}(x_t, h_{t-1}) \rightarrow S$ [Batch, 1024]

PREDICTION VERDICT & EXPLAINABILITY

Dropout(0.4) • Softmax / Sigmoid Probability (REAL vs. FAKE) • Grad-CAM Heatmap
Final Output: Verdict (REAL/FAKE) • Confidence % • Boundary Artifacts

5. TEMPORAL PIPELINE VISUALIZATION

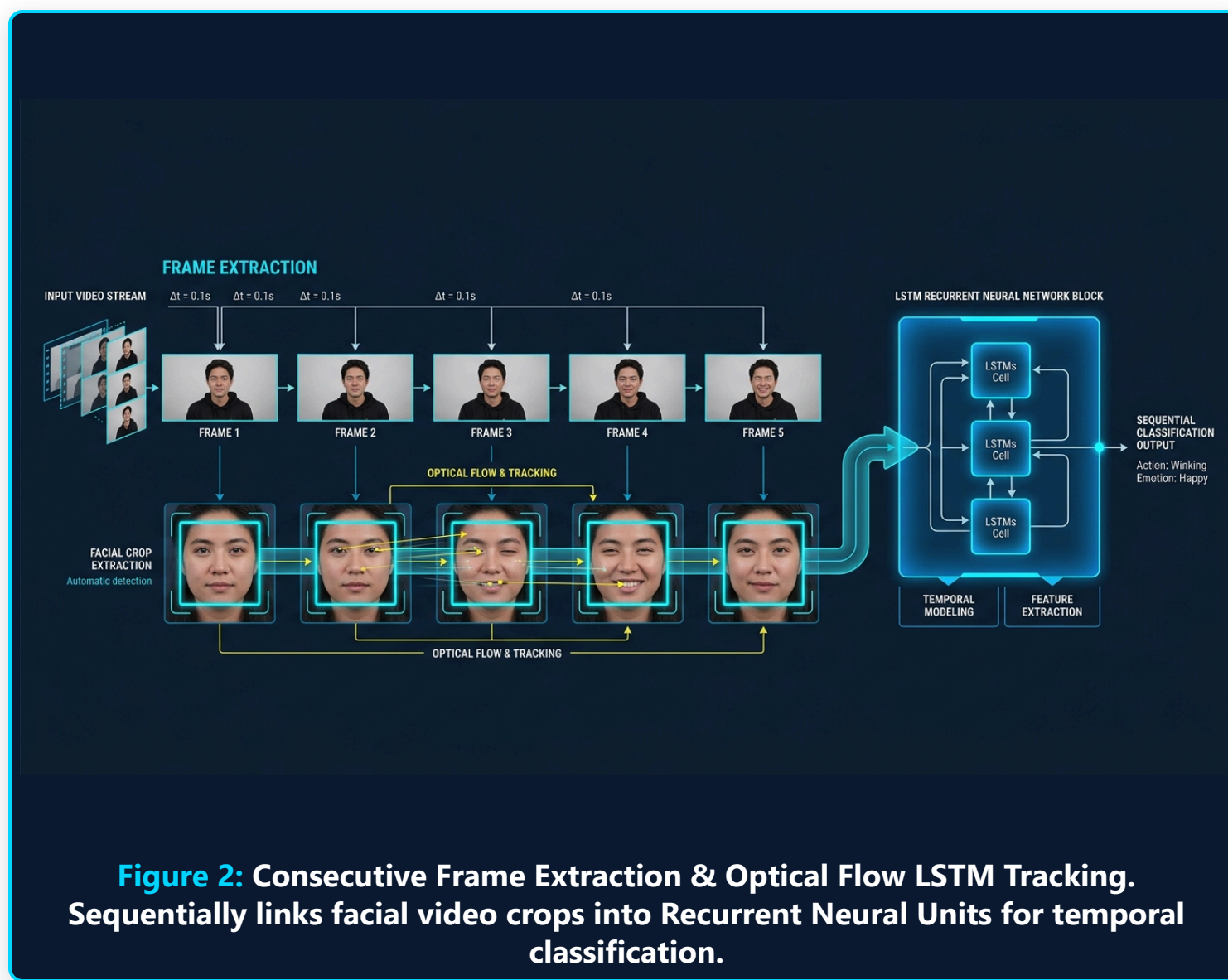


Figure 2: Consecutive Frame Extraction & Optical Flow LSTM Tracking.
Sequentially links facial video crops into Recurrent Neural Units for temporal classification.

- Frame Series:** 20 uniform video crops
- Spatial Feats:** 2048-dim CNN vectors
- LSTM State:** Recurrent temporal flow
- Final Verdict:** Softmax classification

$$L_{\text{total}} = L_{\text{BCE}} + \lambda \cdot ||h_t - h_{t-1}||^2$$

Recurrent hidden states penalize unnatural inter-frame feature variance and flickering jitter.

Window: 1 Sec (20 Frames)

LSTM Units: 512 Hidden Cells

Accuracy Boost: +8.6% vs Static 2D

Single-Frame CNN

Analyzes static pixel patterns only. Misses eye blinking & motion flicker. Vulnerable

Hybrid ResNeXt + LSTM

Tracks 20 sequential frames. Models optical flow & boundary drift. Achieves

10. ABLATION STUDY (PIPELINE GAIN)

ResNeXt-50 (Single Frame Static Baseline)	78.4%
ResNeXt-50 + Frame Averaging (Multi-Frame)	81.2% (+2.8%)
ResNeXt-50 + Gated Recurrent Unit (GRU)	84.5% (+6.1%)
Full Proposed: ResNeXt-50 + Bi-LSTM (20 Frames)	87.0% (+8.6%)

Temporal Impact: Modeling temporal sequence dynamics provides an essential +8.6% accuracy boost over raw spatial features.

13. CONCLUSION & ROADMAP

Conclusion: Spatio-temporal hybrid modeling delivers 87% accuracy, solving single-frame classification limitations.

- Real-Time RTSP:** Direct streaming surveillance camera verification.
- Multimodal Audio-Visual:** Voice-lip synchrony cross-verification.
- Edge AI:** TensorRT deployment for low-latency smartphone screening.

6. BENCHMARK EVALUATION

87%
ACCURACY

0.92
AUC-ROC

23K+
VIDEOS

1.8s
LATENCY

Cross-Dataset Confusion Matrix (87.0% Overall Accuracy)

True: REAL	91.2%	8.8%
True: FAKE	15.2%	84.8%
	Pred: REAL	Pred: FAKE

Figure 3: Cross-Dataset Confusion Matrix (True Real: 91.2%, True Fake: 84.8%).

Precision: 88.4% **Recall:** 85.6% **F1:** 87.0% **Spec:** 89.2%

Validation Insight: Evaluated on compressed DFDC challenge sequences; hybrid temporal fusion retains 84.8% true-positive sensitivity against complex generative reenactments.

7. MODEL HYPERPARAMETERS

Parameter	Engineering Value
Architecture	ResNeXt-50 (32×4d) + LSTM
Frame Horizon	20 Frames / Video (1-Sec Temporal Window)
Crop Resolution	112 × 112 Pixels (RGB Normalized)
Optimization	Adam (lr=1e-5, weight_decay=1e-5)
Loss Objective	BCEWithLogitsLoss (Binary Cross-Entropy)
Regularization	Dropout(p=0.4) + StepLR Schedule (γ=0.1)
Inference Latency	~1.85 Seconds / Video (CUDA GPU)
Explainability	Grad-CAM / Class Activation Maps

TRAINING PROTOCOL & OPTIMIZATION METRICS

Epochs: 25 Full Epochs (Batch=32)	Loss Decay: 0.692 → 0.284 Final
Augmentation: Flip, Affine, Color Jitter	Grad Clipping: Norm threshold = 1.0

8. OFFICIAL PROJECT SHOWCASE & CLOUD DEPLOYMENT



SCAN FOR INTERACTIVE RESEARCH PORTAL

[deepfakedetection-cyan.vercel.app](#)

Scan with any smartphone camera to launch the live detection platform, inspect high-resolution Grad-CAM heatmaps, and test video prediction models!

📱 Mobile & Desktop | 🔍 8K Forensics | ⚡ 24/7 Vercel Edge | 💯 100% Free Access

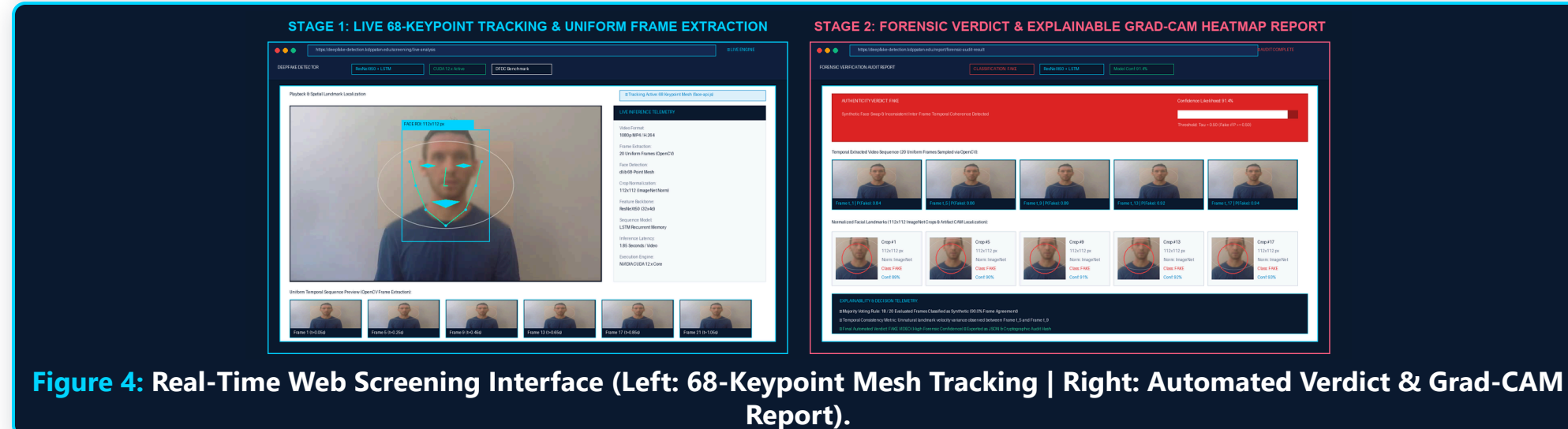


Figure 4: Real-Time Web Screening Interface (Left: 68-Keypoint Mesh Tracking | Right: Automated Verdict & Grad-CAM Report).

INTERACTIVE CLOUD DEMONSTRATION PROTOCOL

- Scan QR Code:** Instantly launch the hosted responsive portal on any mobile device or browser.
- Input Selection:** Select pre-loaded benchmark clips or upload custom video files for analysis.
- Client Landmark Mesh:** Real-time 68-point face tracking executes in-browser via TensorFlow.js.

Deployment: Docker Compose • Cloud: AWS EC2 (Ubuntu 22.04) • GPU: RTX 4090

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11. GENERATOR DEFENSE & DEPLOYMENT

FaceSwap: 89.2% Acc (Boundary edges)	DeepFaceLab: 86.8% Acc (Latent blur)
Face2Face: 88.1% Acc (Reenactment)	NeuralTextures: 84.0% Acc (GAN jitter)

Inference Speed: ~1.85 Seconds / video (NVIDIA CUDA GPU) • Containerized via Docker & Gunicorn for real-time web screening.

14. REFERENCES & ETHICS

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- S. Hochreiter & J. Schmidhuber, "Long Short-Term Memory", Neural Computation.
- A. Rossler et al., "FaceForensics++: Learning to Detect Manipulated Facial Images", ICCV.
- Y. Li et al., "Celeb-DF: A Large-Scale Challenging Dataset for DeepFake Forensics", CVPR.
- R. Selvaraju et al., "Grad-CAM: Visual Explanations from Deep Networks", ICCV.

Ethical AI: Adheres to transparency and privacy guidelines for open forensic benchmarks.